Second-order adjoint methods for frictional parameters in earthquake modeling

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1 Introduction

We study adjoint-based inversion for rate-state frictional parameters in earthquake modeling, as done in [1], and extend the analysis to the second-order adjoint, as done in [2], [3]. To simplify, we start by assuming no changes in fault normal stress. We also consider the initial state Ψ_0 as known (should be easy to generalize later).

2 Derivations

2.1 Forward and adjoint problems

From [1], we have the following systems:
Forward problem

$$\begin{aligned}
\rho \ddot{u}_J &= \partial_I C_{IJKL} \partial_K u_L + Q_J, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\overline{u} &= \overline{u}_0, \quad \dot{\overline{u}} = \overline{v}_0, && \overline{x} \in \Omega, && t = 0, \\
L\overline{u} &= \overline{g}, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
V_n &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\overline{T}^+ &= -\overline{T}^-, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\tau_J &= F_J(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\dot{\Psi} &= G(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Psi &= \Psi_0, && \overline{x} \in \Gamma, && t = 0.
\end{aligned} \tag{1}$$

Adjoint problem

$$\begin{aligned}
\rho \ddot{u}^\dagger_J &= \partial_I C_{IJKL} \partial_K u^\dagger_L + Q^\dagger_J, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\overline{u}^\dagger &= \overline{0}, \quad \dot{\overline{u}}^\dagger = \overline{0}, && \overline{x} \in \Omega, && t = T, \\
L^\dagger \overline{u}^\dagger &= 0, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
V^\dagger_n &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\overline{T}^{\dagger +} &= -\overline{T}^{\dagger -}, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\tau^\dagger_J &= F^\dagger_J(\overline{V}^\dagger, \Psi^\dagger), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
-\dot{\Psi}^\dagger &= G^\dagger(\overline{V}^\dagger, \Psi^\dagger), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Psi^\dagger &= 0, && \overline{x} \in \Gamma, && t = T.
\end{aligned} \tag{2}$$

The adjoint source $Q_J^\dagger$ is the residual (the difference between model output at receivers and data) injected at the receiver locations. The adjoint friction law and state evolution are given by

$$\begin{aligned} F_J^\dagger(\overline{V}^\dagger, \Psi^\dagger) &= \frac{\partial F_I}{\partial V_J} V_I^\dagger + \frac{\partial G}{\partial V_J}\Psi^\dagger, \\ G^\dagger(\overline{V}^\dagger, \Psi^\dagger) &= \frac{\partial F_J}{\partial \Psi} V_J^\dagger + \frac{\partial G}{\partial \Psi}\Psi^\dagger. \end{aligned} \tag{3}$$

The adjoint problem is linear, with coefficients that depend on the forward solution. One way to view the derivation is that we linearize the forward problem around the forward solution and then take the adjoint.

2.2 Second-order Lagrangian

The cost functional for the second-order problem is constructed by convolving the gradient (5.9 from [1]) with a fixed model perturbation in the parameter p_1. We here denote the perturbation Δp_1.

$$\mathcal{F}_2 = -\left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1\right)_{\Gamma\times\mathcal{T}}. \tag{4}$$

We construct the Lagrangian by adding the forward and adjoint problems [3]. The forward problem (FP) is multiplied by Lagrange multipliers $\Delta u_j^\dagger, \Delta\Psi^\dagger$ (the second-order adjoint solution) and the adjoint problem (AP) by $\Delta u_j, \Delta\Psi$ (the second-order forward solution).

$$\begin{aligned} \mathcal{L}_2 = &-\left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\ &+ \left(\Delta\dot{u}_j^\dagger, \rho\ddot{u}_j - \partial_i C_{ijkl}\partial_k u_l - Q_j\right)_{\Omega\times\mathcal{T}} + \left(\Delta\Psi^\dagger, \dot{\Psi} - G\right)_{\Gamma\times\mathcal{T}} \\ &+ \left(\Delta\dot{u}_j, \rho\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k u_l^\dagger - Q_j^\dagger\right)_{\Omega\times\mathcal{T}} + \left(\Delta\Psi, -\dot{\Psi}^\dagger - G^\dagger\right)_{\Gamma\times\mathcal{T}} \end{aligned} \tag{5}$$

Varying the second-order variables straightforwardly leads to the forward and adjoint problems. Before varying the forward and adjoint variables, we use integration by parts to rewrite the Lagrangian. From the derivations in [1], we know that

$$\begin{aligned} &\left(\dot{u}_j^\dagger, \rho\ddot{u}_j - \partial_i C_{ijkl}\partial_k u_l - Q_j\right)_{\Omega\times\mathcal{T}} + \left(\Psi^\dagger, \dot{\Psi} - G\right)_{\Gamma\times\mathcal{T}} \\ = &-\left(\rho\dot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k u_l, \dot{u}_j\right)_{\Omega\times\mathcal{T}} - \left(\dot{u}_j^\dagger, Q_j\right)_{\Omega\times\mathcal{T}} \\ &- \left(V_j^\dagger, F_j\right)_{\Gamma\times\mathcal{T}} + \left(F_j^\dagger, V_j\right)_{\Gamma\times\mathcal{T}} + \underbrace{\left(V_n^\dagger, \sigma_n\right)_{\Gamma\times\mathcal{T}}}_{\text{Ignore, assume no change in } \sigma_n} \\ &- \left(\Psi^\dagger, \Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Psi^\dagger, G\right)_{\Gamma\times\mathcal{T}} + \underbrace{\left(\Psi_0^\dagger, \Psi_0\right)_\Gamma}_{\text{Ignore for now}} \end{aligned} \tag{6}$$

The derivations assume initial and terminal conditions, correct adjoint boundary conditions on exterior boundaries, and continuity of forward and adjoint traction across the fault. Using (6) on the terms in (5), and thereby assuming homogeneous initial/terminal

conditions, allows us to rewrite $\mathcal{L}_2$ as

$$\begin{aligned}\mathcal{L}_2 = &-\left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k \Delta u_l^\dagger, \dot{u}_j\right)_{\Omega\times\mathcal{T}} - \left(\Delta\dot{u}_j^\dagger, Q_j\right)_{\Omega\times\mathcal{T}} \\ &-\left(\Delta V_j^\dagger, F_j\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, V_j\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\dot{\Psi}^\dagger, \Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, G\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\rho\Delta\ddot{u}_j - \partial_i C_{ijkl}\partial_k \Delta u_l, \dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}} - \left(\Delta\dot{u}_j, Q_j^\dagger\right)_{\Omega\times\mathcal{T}} \\ &-\left(\Delta V_j, F_j^\dagger\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j, V_j^\dagger\right)_{\Gamma\times\mathcal{T}} + \left(\Delta\dot{\Psi}, \Psi^\dagger\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, G^\dagger\right)_{\Gamma\times\mathcal{T}}\end{aligned} \tag{7}$$

As expected, this closely resembles equation (5.21) in [1]. Note that we disregard the Ψ_0-terms to make it more concise.

For several parameter fields $\mathbf{p} = (p^{(1)}, \ldots, p^{(N)})$, products such as $\frac{\partial F}{\partial p_1}\Delta p_1$ are understood as

$$\frac{\partial F}{\partial p_1}\Delta p_1 = \sum_{k=1}^{N} \frac{\partial F}{\partial p_1^{(k)}}\,\Delta p_1^{(k)}, \tag{8}$$

i.e. the directional derivative of F in the direction Δp_1. This follows the multi-parameter formulation of [2], see their Eqs. (27), (57) and (60). Maybe this is not clear with the current notation?

2.3 Variations

Varying the parameter p_2 leads to

$$\begin{aligned}\partial_{p_2}\mathcal{L}_2 = &-\left(V_j^\dagger \frac{\partial^2 F_j}{\partial p_1 \partial p_2} + \Psi^\dagger \frac{\partial^2 G}{\partial p_1 \partial p_2}, \Delta p_1 \delta p_2\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\Delta V_j^\dagger, \frac{\partial F_j}{\partial p_2}\delta p_2\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, \frac{\partial G}{\partial p_2}\delta p_2\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\Delta V_j, \frac{\partial F_j^\dagger}{\partial p_2}\delta p_2\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, \frac{\partial G^\dagger}{\partial p_2}\delta p_2\right)_{\Gamma\times\mathcal{T}}\end{aligned} \tag{9}$$

The partial derivatives of the adjoint friction law and state evolution functions are

$$\begin{aligned}\frac{\partial F_j^\dagger}{\partial p} &= \frac{\partial^2 F_i}{\partial V_j \partial p} V_i^\dagger + \frac{\partial^2 G}{\partial V_j \partial p}\Psi^\dagger, \\ \frac{\partial G^\dagger}{\partial p} &= \frac{\partial^2 F_j}{\partial \Psi \partial p} V_j^\dagger + \frac{\partial^2 G}{\partial \Psi \partial p}\Psi^\dagger,\end{aligned} \tag{10}$$

which leads to

$$\begin{aligned}\partial_{p_2}\mathcal{L}_2 = &-\left(V_j^\dagger \frac{\partial^2 F_j}{\partial p_1 \partial p_2} + \Psi^\dagger \frac{\partial^2 G}{\partial p_1 \partial p_2}, \Delta p_1 \delta p_2\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\Delta V_j^\dagger \frac{\partial F_j}{\partial p_2}, \delta p_2\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger \frac{\partial G}{\partial p_2}, \delta p_2\right)_{\Gamma\times\mathcal{T}} \\ &-\left(V_j^\dagger \left[\frac{\partial^2 F_j}{\partial V_i \partial p_2}\Delta V_i + \frac{\partial^2 F_j}{\partial \Psi \partial p_2}\Delta\Psi\right], \delta p_2\right)_{\Gamma\times\mathcal{T}} \\ &-\left(\Psi^\dagger \left[\frac{\partial^2 G}{\partial V_i \partial p_2}\Delta V_i + \frac{\partial^2 G}{\partial \Psi \partial p_2}\Delta\Psi\right], \delta p_2\right)_{\Gamma\times\mathcal{T}}.\end{aligned} \tag{11}$$

Now, we seek the second-order forward problem and second-order adjoint problem. We start by varying the Lagrangian (7) with respect to the adjoint solution, which will lead to the second-order forward problem. Note that we are essentially taking the adjoint of the adjoint problem here, since we are repeating the steps that led to the first-order adjoint equations, but with the adjoint solution playing the role of the forward solution and the second-order forward solution playing the role of the adjoint solution. The variation with respect to the adjoint solution reads

$$\begin{aligned}\delta\mathcal{L}_2 = &-\left(\delta\Psi^\dagger\frac{\partial G}{\partial p_1}+\delta V_j^\dagger\frac{\partial F_j}{\partial p_1},\Delta p_1\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\Delta V_j,\delta F_j^\dagger\right)_{\Gamma\times\mathcal{T}}+\left(\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &+\left(\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}-\left(\Delta\Psi,\delta G^\dagger\right)_{\Gamma\times\mathcal{T}}.\end{aligned} \tag{12}$$

The required variations of $F_j^\dagger$ and $G^\dagger$ are

$$\begin{aligned}\delta F_j^\dagger &= \frac{\partial F_i}{\partial V_j}\delta V_i^\dagger+\frac{\partial G}{\partial V_j}\delta\Psi^\dagger,\\ \delta G^\dagger &= \frac{\partial F_j}{\partial\Psi}\delta V_j^\dagger+\frac{\partial G}{\partial\Psi}\delta\Psi^\dagger.\end{aligned} \tag{13}$$

Substituting in (12) and rearranging terms yields

$$\begin{aligned}\delta\mathcal{L}_2 = &-\left(\delta\Psi^\dagger\frac{\partial G}{\partial p_1}+\delta V_j^\dagger\frac{\partial F_j}{\partial p_1},\Delta p_1\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\Delta V_j,\frac{\partial F_i}{\partial V_j}\delta V_i^\dagger+\frac{\partial G}{\partial V_j}\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}+\left(\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &+\left(\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}-\left(\Delta\Psi,\frac{\partial F_j}{\partial\Psi}\delta V_j^\dagger+\frac{\partial G}{\partial\Psi}\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}\\ = &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\frac{\partial F_j}{\partial p_1}\Delta p_1+\Delta V_i\frac{\partial F_j}{\partial V_i}+\Delta\Psi\frac{\partial F_j}{\partial\Psi}-\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\frac{\partial G}{\partial p_1}\Delta p_1+\Delta V_j\frac{\partial G}{\partial V_j}+\Delta\Psi\frac{\partial G}{\partial\Psi}-\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}\end{aligned} \tag{14}$$

The variation is zero if the second-order solution solves the following problem.

Second-order forward problem

$$\begin{aligned}
\rho\Delta\ddot{u}_J &= \partial_I C_{IJKL}\partial_K \Delta u_L, && \overline{x}\in\Omega, && t\in\mathcal{T},\\
\Delta\overline{u} &= \overline{0}, \quad \Delta\dot{\overline{u}} = \overline{0}, && \overline{x}\in\Omega, && t=0,\\
L\Delta\overline{u} &= \overline{0}, && \overline{x}\in\partial\Omega, && t\in\mathcal{T},\\
\Delta V_n &= 0, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\overline{T}^{+} &= -\Delta\overline{T}^{-}, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\tau_J &= \Delta F_J, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\dot{\Psi} &= \Delta G, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\Psi &= 0, && \overline{x}\in\Gamma, && t=0.
\end{aligned} \tag{15}$$

where

$$\begin{aligned}
\Delta F_J &= \frac{\partial G^\dagger}{\partial V_J^\dagger}\Delta\Psi + \frac{\partial F_I^\dagger}{\partial V_J^\dagger}\Delta V_I + \frac{\partial F_J}{\partial p_1}\Delta p_1\\
&= \frac{\partial F_J}{\partial \Psi}\Delta\Psi + \frac{\partial F_J}{\partial V_I}\Delta V_I + \frac{\partial F_J}{\partial p_1}\Delta p_1\\
\Delta G &= \frac{\partial G^\dagger}{\partial \Psi^\dagger}\Delta\Psi + \frac{\partial F_J^\dagger}{\partial \Psi^\dagger}\Delta V_J + \frac{\partial G}{\partial p_1}\Delta p_1\\
&= \frac{\partial G}{\partial \Psi}\Delta\Psi + \frac{\partial G}{\partial V_J}\Delta V_J + \frac{\partial G}{\partial p_1}\Delta p_1
\end{aligned} \tag{16}$$

The second-order forward problem is the linearized forward problem, but with different source terms. This is consistent with [3]. It is the adjoint of the adjoint problem, which in turn was the adjoint of the linearized forward problem, so after taking the adjoint twice we are back to the linearized forward problem. Unlike the adjoint problem, there is no wavefield source at receiver locations. Instead, the source terms appear in the friction law and state evolution. They depend on the forward solution and the model perturbation Δp_1. There is no dependence on the adjoint solution. The second-order forward solution appears to be the (first-order) change in the forward solution that the model perturbation Δp_1 would cause.

To derive the second-order adjoint problem, we vary the forward solution, yielding

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\Psi^\dagger\left(\frac{\partial^2 G}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 G}{\partial p_1\partial\Psi}\delta\Psi\right) + V_j^\dagger\left(\frac{\partial^2 F_j}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 F_j}{\partial p_1\partial\Psi}\delta\Psi\right), \Delta p_1\right)_{\Gamma\times\mathcal{T}}\\
&-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k\Delta u_l^\dagger, \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}}\\
&-\left(\Delta V_j^\dagger, \delta F_j\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, \delta G\right)_{\Gamma\times\mathcal{T}}\\
&-\left(\Delta\dot{u}_j, \delta Q_j^\dagger\right)_{\Omega\times\mathcal{T}}\\
&-\left(\Delta V_j, \delta F_j^\dagger\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, \delta G^\dagger\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{17}$$

There are more terms than when varying the adjoint solution, because $F_j^\dagger$, $G^\dagger$ and $Q_j^\dagger$ all depend on the forward solution, whereas F_j, G and Q_j are independent of the adjoint solution. Recall that, for a velocity misfit, the adjoint source is

$$Q_j^\dagger(t) = \sum_{k=1}^{N_{rec}} -(\dot{u}_j(\overline{x}_r^{(k)}, t) - d_j(t))\hat{\delta}(\overline{x} - \overline{x}_r^{(k)}) \tag{18}$$

The required variations are (need better notation for different variations)

$$\begin{aligned}
\delta F_j &= \frac{\partial F_j}{\partial V_i}\delta V_i + \frac{\partial F_j}{\partial \Psi}\delta\Psi \\
\delta G &= \frac{\partial G}{\partial V_j}\delta V_j + \frac{\partial G}{\partial \Psi}\delta\Psi \\
\delta F_j^\dagger &= \frac{\partial F_j^\dagger}{\partial V_i}\delta V_i + \frac{\partial F_j^\dagger}{\partial \Psi}\delta\Psi \\
\delta G^\dagger &= \frac{\partial G^\dagger}{\partial V_i}\delta V_i + \frac{\partial G^\dagger}{\partial \Psi}\delta\Psi \\
\delta Q_j^\dagger &= \sum_{k=1}^{N_{rec}} -\delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)})
\end{aligned} \tag{19}$$

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\Psi^\dagger\left(\frac{\partial^2 G}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 G}{\partial p_1\partial\Psi}\delta\Psi\right) + V_j^\dagger\left(\frac{\partial^2 F_j}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 F_j}{\partial p_1\partial\Psi}\delta\Psi\right), \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k\Delta u_l^\dagger - \sum_{k=1}^{N_{rec}}\Delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)}), \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}} \\
&-\left(\Delta V_j^\dagger, \frac{\partial F_j}{\partial V_i}\delta V_i + \frac{\partial F_j}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, \frac{\partial G}{\partial V_j}\delta V_j + \frac{\partial G}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\Delta V_j, \frac{\partial F_j^\dagger}{\partial V_i}\delta V_i + \frac{\partial F_j^\dagger}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, \frac{\partial G^\dagger}{\partial V_i}\delta V_i + \frac{\partial G^\dagger}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{20}$$

Gathering terms leads to

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k\Delta u_l^\dagger - \sum_{k=1}^{N_{rec}}\Delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)}), \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}} \\
&-\left(\left(\Psi^\dagger\frac{\partial^2 G}{\partial p_1\partial\Psi} + V_j^\dagger\frac{\partial^2 F_j}{\partial p_1\partial\Psi}\right)\Delta p_1 + \frac{\partial F_j}{\partial\Psi}\Delta V_j^\dagger + \frac{\partial G}{\partial\Psi}\Delta\Psi^\dagger + \frac{\partial F_j^\dagger}{\partial\Psi}\Delta V_j + \frac{\partial G^\dagger}{\partial\Psi}\Delta\Psi + \Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\left(\Psi^\dagger\frac{\partial^2 G}{\partial p_1\partial V_j} + V_i^\dagger\frac{\partial^2 F_i}{\partial p_1\partial V_j}\right)\Delta p_1 + \frac{\partial F_i}{\partial V_j}\Delta V_i^\dagger + \frac{\partial G}{\partial V_j}\Delta\Psi^\dagger + \frac{\partial F_i^\dagger}{\partial V_j}\Delta V_i + \frac{\partial G^\dagger}{\partial V_j}\Delta\Psi - \Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{21}$$

Note that the variations of $F_j^\dagger$ and $G^\dagger$ can be unpacked further, if desired:

$$\delta F_j^\dagger = \underbrace{\left(\frac{\partial^2 F_k}{\partial V_j\partial V_i}V_k^\dagger + \frac{\partial^2 G}{\partial V_j\partial V_i}\Psi^\dagger\right)}_{\frac{\partial F_j^\dagger}{\partial V_i}}\delta V_i + \underbrace{\left(\frac{\partial^2 F_k}{\partial V_j\partial\Psi}V_k^\dagger + \frac{\partial^2 G}{\partial V_j\partial\Psi}\Psi^\dagger\right)}_{\frac{\partial F_j^\dagger}{\partial\Psi}}\delta\Psi, \tag{22}$$

$$\delta G^\dagger = \underbrace{\left(\frac{\partial^2 F_j}{\partial\Psi\partial V_i}V_j^\dagger + \frac{\partial G^2}{\partial\Psi\partial V_i}\Psi^\dagger\right)}_{\frac{\partial G^\dagger}{\partial V_i}}\delta V_i + \underbrace{\left(\frac{\partial^2 F_j}{\partial^2\Psi}V_j^\dagger + \frac{\partial^2 G}{\partial^2\Psi}\Psi^\dagger\right)}_{\frac{\partial G^\dagger}{\partial\Psi}}\delta\Psi. \tag{23}$$

The variation vanishes if the second-order adjoint solution satisfies the problem below.

Second-order adjoint problem

$$
\begin{aligned}
\rho \Delta \ddot{u}_J^\dagger &= \partial_I C_{IJKL} \partial_K \Delta u_L^\dagger + \Delta Q_J^\dagger, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\Delta \overline{u}^\dagger &= \overline{0}, \quad \Delta \dot{\overline{u}}^\dagger = \overline{0}, && \overline{x} \in \Omega, && t = T, \\
L^\dagger \Delta \overline{u}^\dagger &= 0, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
\Delta V_n^\dagger &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Delta \overline{T}^{\dagger +} &= -\Delta \overline{T}^{\dagger -}, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Delta \tau_J^\dagger &= \Delta F_J^\dagger, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
-\Delta \dot{\Psi}^\dagger &= \Delta G^\dagger, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Delta \Psi^\dagger &= 0, && \overline{x} \in \Gamma, && t = T.
\end{aligned}
\tag{24}
$$

where

$$
\begin{aligned}
\Delta F_j^\dagger &= \frac{\partial G^\dagger}{\partial V_j} \Delta \Psi + \frac{\partial F_i^\dagger}{\partial V_j} \Delta V_i + \frac{\partial G}{\partial V_j} \Delta \Psi^\dagger + \frac{\partial F_i}{\partial V_j} \Delta V_i^\dagger \\
&\quad + \Delta p_1 \left(\Psi^\dagger \frac{\partial^2 G}{\partial V_j \partial p_1} + V_i^\dagger \frac{\partial^2 F_i}{\partial V_j \partial p_1} \right), \\
\Delta G^\dagger &= \frac{\partial G^\dagger}{\partial \Psi} \Delta \Psi + \frac{\partial F_j^\dagger}{\partial \Psi} \Delta V_j + \frac{\partial G}{\partial \Psi} \Delta \Psi^\dagger + \frac{\partial F_j}{\partial \Psi} \Delta V_j^\dagger \\
&\quad + \Delta p_1 \left(\Psi^\dagger \frac{\partial^2 G}{\partial \Psi \partial p_1} + V_j^\dagger \frac{\partial^2 F_j}{\partial \Psi \partial p_1} \right)
\end{aligned}
\tag{25}
$$

$$
\Delta Q_j^\dagger = \sum_{k=1}^{N_{rec}} \Delta \dot{u}_j \hat{\delta}(\overline{x} - \overline{x}_r^{(k)}) \tag{26}
$$

Just like the adjoint problem, the second-order adjoint problem is solved in reversed time. It is identical to the adjoint problem up to source terms. It has both a wavefield source $\Delta Q_j^\dagger$ and source terms in the friction law and state evolution. The wavefield source $\Delta Q_j^\dagger$ contains the second order forward wavefield $\Delta \dot{u}_j$ injected at the receivers, similar to how the forward wavefield is injected in the adjoint problem. The coefficients in the friction law and state evolution equation depend only on the forward solution (just like for the adjoint problem), but the source terms also depend on the adjoint solution and the model perturbation. Thus the second-order adjoint problem depends on all three previous problems and must be solved last.

Summary: To compute the Hessian applied to a perturbation Δp_1, we need to solve the FP, AP, second-order FP and second-order AP, and then evaluate (11).

Note that we assumed no filtering for simplicity, i.e. $W[x] = x$, which makes the variation of $Q_j^\dagger$ easy to compute.

3 Interpretation

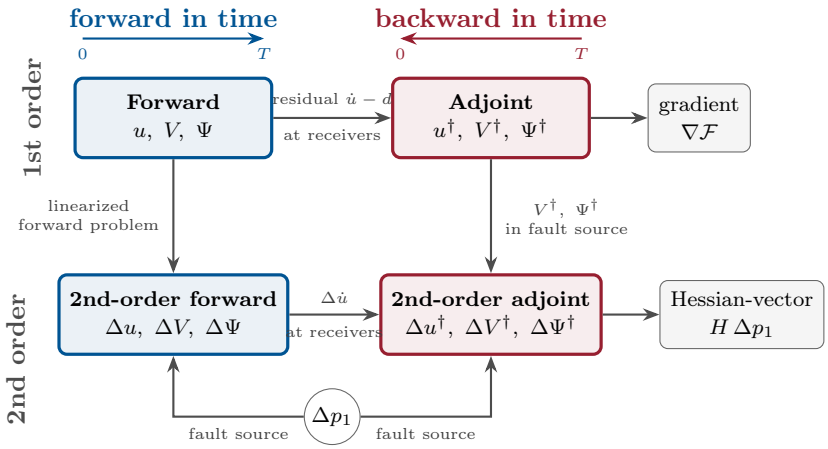


Figure 1: Procedure for one Hessian-vector product $H\,\Delta p_1$, which needs four solves: forward, adjoint, second-order forward and second-order adjoint.

Our variables are u, V and Ψ. The adjoint problem is driven by an adjoint source which is simply the residual at receivers between our model output $\dot{u}$ and data d.

The second order forward problem is the linearized forward problem (linearized in the sense that the nonlinear friction law is linearized). The dependence is from the forward problem (shows up in friction and state laws), and the source is from the model perturbation. The second order adjoint problem is also driven by the model pertubation. It depends on the forward problem via the friction and state laws, and the second order adjoint problem

4 Results

4.1 Convergence

We follow part of what was done in [1]. The hessian vector product can be computed with a forward finite difference approximation,

$$\left(\mathbf{D}_+(\nabla \boldsymbol{\mathcal{F}}^\mathsf{T} \boldsymbol{\Delta p_1})(h)\right)_i = \frac{\left(\nabla \boldsymbol{\mathcal{F}}(\mathbf{p} + h\,\mathbf{e}_i) - \nabla \boldsymbol{\mathcal{F}}(\mathbf{p})\right)^\mathsf{T} \boldsymbol{\Delta p_1}}{h}. \tag{27}$$

A few words regarding this expression. First of all, note that this mimics the FD gradient computation, where we simply use $\nabla \boldsymbol{\mathcal{F}}$ instead of $\boldsymbol{\mathcal{F}}$. To speed up the computation (linear time complexity instead of quadratic), we can compute the gradient of the misfit $\nabla \boldsymbol{\mathcal{F}}$ with the adjoint method, given that we are sure of the adjoint method's correctness. Note also that Δp_1 is fixed and chosen. This is the direction the Hessian is applied to, yielding $H\Delta p_1$. δp_2 is an arbitrary variation, i.e. test direction to read of the result. Discretely, it picks out a component ($\delta p_1 = e_i$). So, Δp_1 is the change we make to the fault and δp_2 is where we look at the effect. The error E is proportional to the step size, $E \propto \mathcal{O}(\Delta p)$. Therefore, we get the following expected behaviour. See table 1 for setup details.

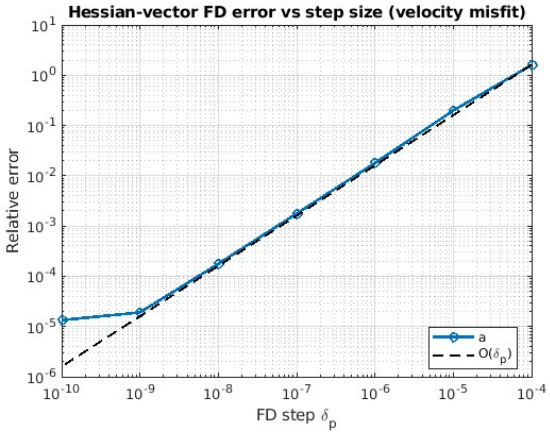


Table 1: Setup of the 2D Hessian-vector verification test.

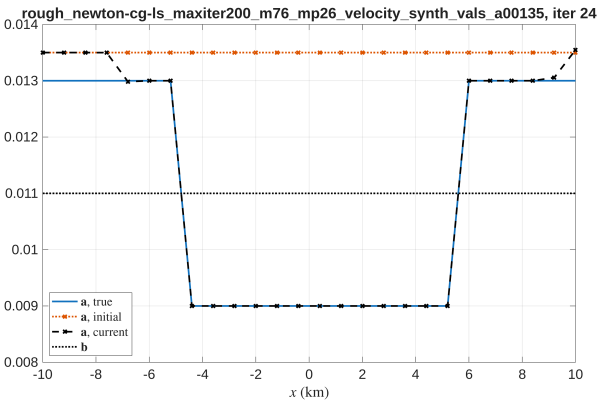
	Quantity	Symbol	Value
Domain	Domain	Ω	$[-10, 10] \times [-10, 10]$
	Fault	Γ	fractal profile, $x \in [-10, 10]$
	Outer boundary condition	–	outflow (all four sides)
	Final time	T	4
Discretization	Grid points per block side	m	26
	Parameter grid points	m_p	26
	SBP operators	–	boundary-optimized, order 6
	Time stepper	–	explicit RK4, fixed step
	Time step	k	5×10^{-3}
Material	Shear wave speed	c_s	3.464 km/s
	Density	ρ	2.67 g/cm^3
	Shear modulus	$\mu = \rho c_s^2$	32.04 GPa
Friction	Friction law	F	$\sigma_0 a \operatorname{asinh}\left(\frac{V}{2V_0} e^{\Psi/a}\right) - \tau_0$
	State evolution	G	slip law
	Direct effect (true)	a	0.009 (VW, $x \in [-5, 6]$), 0.013 (VS)
	Direct effect (initial guess)	a_0	$1.1\,a$
	Evolution effect	b	0.011
	Critical slip distance	D_c	0.2 (VW), 1 (VS)
	Reference friction coefficient	f_0	0.6
	Reference slip rate	V_0	10^{-6}
	Normal stress	σ_0	120 MPa
	Background shear stress	τ_0	72 MPa
Initial data & loading	Initial slip rate	V_{init}	10^{-12}
	Initial state	Ψ_0	0.7243
	Loading	$\tau_L(x)$	$A \exp\left(-(x - x_0)^2/(2s^2)\right)$
	Loading parameters	$A,\ x_0,\ s$	25, 3, 2
Data & misfit	Receivers	N_r	88, spacing 2, off-fault ($\lvert y\rvert \geq 2$ or $\lvert x\rvert \geq 7$)
	Misfit	$\mathcal{F}$	velocity, L^2 in time
	Data filtering	–	none
Hessian test	Inversion parameter	$\mathbf{p}$	a (nodal values on Γ)
	Direction	$\boldsymbol{\varepsilon}$	$\mathbf{e}_{17}$ ($x \approx 3.0$)
	FD step (single test)	Δp	10^{-9}
	FD step (sweep)	Δp	$10^{-4}, 10^{-5}, \ldots, 10^{-10}$

4.2 Coarse inversion

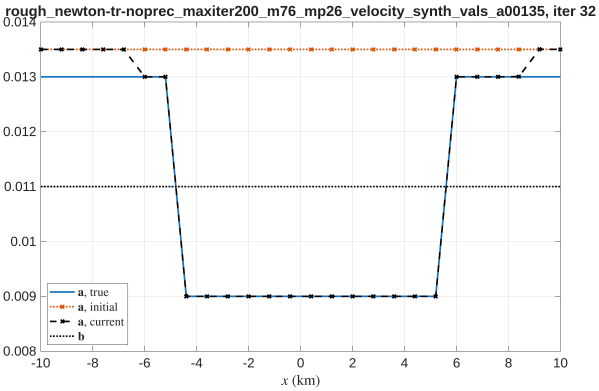
We perform an inverse crime for a variety of solvers. We test fminunc-lbfgs, fmincon-CG (interior-point), newton-tr (L-BFGS preconditioning), newton-tr-noprec (i.e. plain Steihaug CG), fminunc-tr and newton-cg-ls (Line-search Newton-CG with Armijo backtracking by M. Heinkenschloss). Did set lb = zeros(size(x0)) for newton-tr, probably OK since it clearly still works well. We now run with $m = 76$ and $m_p = 26$.

Table 2: Final results of the $m = 76$ inversions. L-BFGS stopped at the iteration limit (200) and `fminunc` trust-region on its step tolerance; all other methods converged. Relative error is $\|\mathbf{a} - \mathbf{a}_{\text{true}}\|_2 / \|\mathbf{a}_{\text{true}}\|_2$ over all $m_p = 26$ parameter nodes. N.P. = no preconditioner

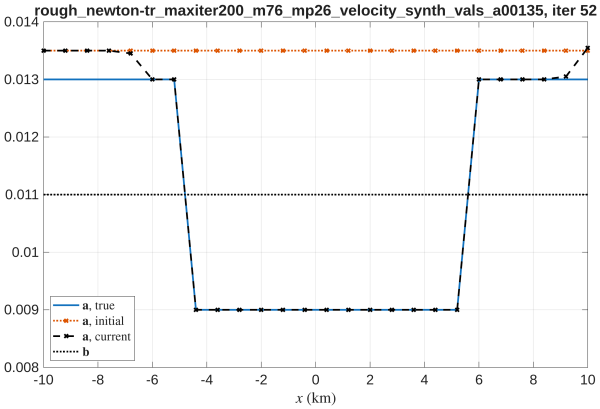
Method	Final misfit f	Rel. error in a	Wall time (h)	Iterations
Line-search Newton-CG	8.72×10^{-7}	0.0200	1.2	24
Trust-region Newton-CG, n.p.	8.74×10^{-7}	0.0232	1.4	32
Trust-region Newton-CG	8.72×10^{-7}	0.0215	1.4	52
Interior-point CG (`fmincon`)	8.72×10^{-7}	0.0200	2.9	31
Trust-region (`fminunc`)	3.57×10^{-3}	0.0368	2.8	68
L-BFGS (`fmincon`)	1.36	0.1254	2.5	200



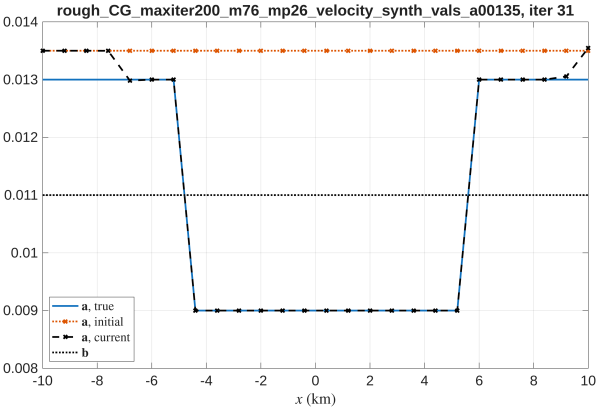
(a) Line-search Newton-CG (24 iterations)



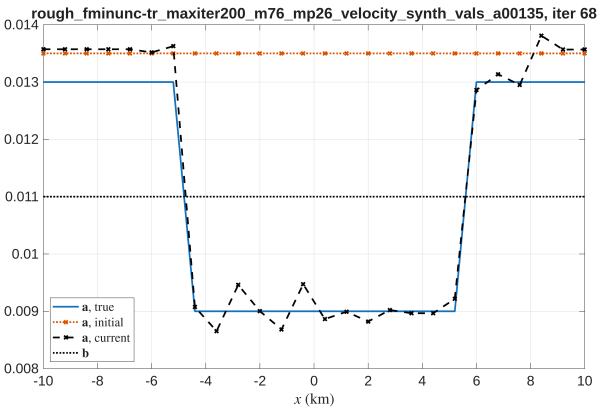
(b) Trust-region Newton-CG, no preconditioner (32 iterations)



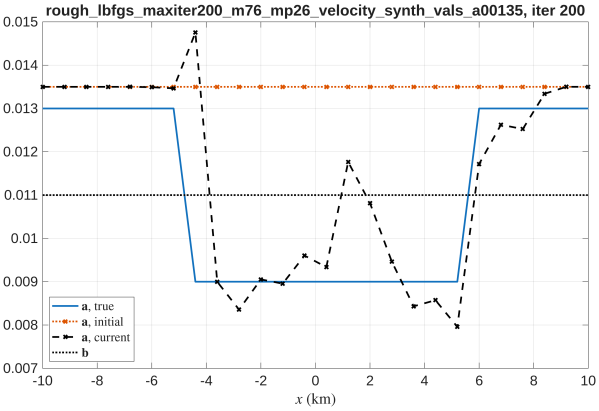
(c) Trust-region Newton-CG (52 iterations)



(d) Interior-point CG, `fmincon` (31 iterations)



(e) Trust-region, `fminunc` (68 iterations)

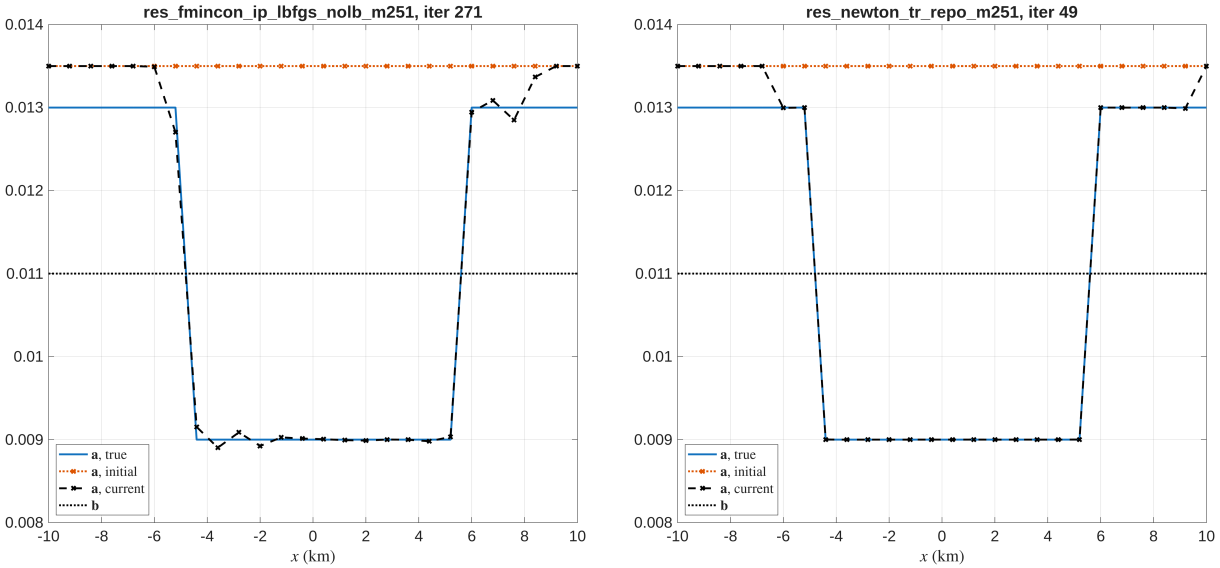


(f) L-BFGS, `fmincon` (200 iterations)

Figure 2: Inverted **a** *at the final iteration of each solver,* $m = 76$*,* $m_p = 26$*.*

4.3 Fine inversion

We see that when using a fine grid, CG does not beat L-BFGS as hard, but still a big difference in terms of inversion error and computation time.



(a) 271 iterations *(b) 49 iterations*

Figure 3: Comparing L-BFGS and trust-region newton CG

We run with $m = 251$, $m_p = 26$, SBP order 8, fixed time stepping with RK4.

Table 3: Final results of the $m = 251$ inversions. Cost is in units of one gradient evaluation. L-BFGS stopped at its budget; Newton-CG converged. Relative error is $\|\mathbf{a} - \mathbf{a}_{\text{true}}\|_2/\|\mathbf{a}_{\text{true}}\|_2$ over all $m_p = 26$ parameter nodes.

Method	Final misfit f	Rel. error in a	Cost	Wall time (h)	Iterations
Trust-region Newton-CG	2.36×10^{-5}	0.0215	166	5.9	49
L-BFGS (`fmincon`)	1.46×10^{-4}	0.0266	401	13.3	271

5 Inversion using high-resolution synthetic data

We now invert data generated on a fine grid ($m = 1001$) on coarser grids ($m = 76$, 151 and 251), with $m_p = 26$ unless stated otherwise. The data then contain a discretization error that no $\mathbf{a}$ on the coarse grid can reproduce, and the minimum misfit is no longer zero.

Without regularization, both solvers reach the minimum misfit $F \approx 0.73$ in about the same time (Fig. 4). Converging further only fits the discretization error, so the converged Newton-CG solution is worse than the L-BFGS one (relative error 0.29 against 0.12 on the resolved nodes), and its best iterate is only reached if the iteration is stopped at the right time. We therefore minimize $J = F + R$ with

$$R_{\text{grad}}(\mathbf{p}) = \frac{\beta}{2}\|\mathbf{D}\mathbf{p}\|_2^2 \quad \text{or} \quad R_{\text{TV}}(\mathbf{p}) = \beta \sum_{i=1}^{m_p-1} \phi_\varepsilon\big((\mathbf{D}\mathbf{p})_i\big), \qquad \phi_\varepsilon(x) = \sqrt{x^2+\varepsilon^2} - \varepsilon, \quad (28)$$

where $(\mathbf{D}\mathbf{p})_i = p_{i+1} - p_i$. The minimizer of J does not depend on when the iteration is stopped, so the solvers can be compared by the wall time to reach it. We use the time $t_{10^{-4}}$ to reach $(J - J^*)/(J_0 - J^*) = 10^{-4}$, where J^* is the smallest objective reached by any of the compared runs. At this level $\mathbf{a}$ is within about 1% of its final value.

Table 4 and Fig. 5 show the cases where Newton-CG is faster. With tikhonov-grad, Newton-CG reaches the minimizer 1.6 times faster for $\beta = 10^3$, and for $\beta = 10^4$ (the smallest error in $\mathbf{a}$ of all runs) L-BFGS does not reach the level within its 200 iterations, while Newton-CG takes 0.35 h. With TV, Newton-CG is 1.5 times faster both at $m = 76$ and at $m = 251$. Both solvers find the same $\mathbf{a}$ on the resolved nodes (Fig. 6).

Newton-CG can stall at the constant initial guess. There $\mathbf{Dp} = 0$, and the Hessian of R_{TV} is $(\beta/\varepsilon)\,\mathbf{D}^\top\mathbf{D}$, so Newton-CG makes almost no progress until it leaves this stiff region (the flat start in Fig. 5). The stall grows with β/ε (Fig. 7): for $\beta = 100$ it lasts 0.5 h with $\varepsilon = 10^{-3}$, 5.5 h with $\varepsilon = 10^{-4}$, and Newton-CG makes no progress at all with $\varepsilon = 10^{-5}$ or $\varepsilon = 0$. L-BFGS is insensitive to ε. The regularization is not the only cause: at $m = 151$ with tikhonov-grad and $\beta = 10^3$, where $\beta\,\mathbf{D}^\top\mathbf{D}$ is mild, Newton-CG does not leave the initial guess in 16 iterations (3.5 h, Table 6).

Hybrid L-BFGS/Newton-CG. Starting Newton-CG from an L-BFGS iterate avoids the stall (Table 5, Fig. 8). We start Newton-CG from iteration $k = 20$ or $k = 40$ of the L-BFGS run, where $(J - J^*)/(J_0 - J^*) \approx 10^{-1}$ and $2 \cdot 10^{-2}$, and count the L-BFGS time. With $k = 20$, the hybrid reaches the level after 0.98 h for TV with $\beta = 10$, where Newton-CG alone (1.89 h) is slower than L-BFGS, and after 1.39 h for $\beta = 100$, against 6.47 h for Newton-CG alone. L-BFGS reaches the level in neither case within 200 iterations. For $\beta = 100$, however, the hybrid is only about as fast as L-BFGS until L-BFGS stops (1.35 h against 1.46 h to the final L-BFGS objective); its gain is that it continues to J^*. When Newton-CG does not stall (tikhonov-grad), starting it directly is fastest, and switching later ($k = 40$) is slower in all cases. All runs end at the same J^* and $\mathbf{a}$.

In the inverse-crime setting the hybrid does not help. At $m = 76$, Newton-CG started from L-BFGS iteration 30 or 60 stops at the same misfit as L-BFGS ($f \approx 1.36$, relative error 0.19), whereas Newton-CG started from the initial guess reaches $f \approx 10^{-6}$ (error 0.02). L-BFGS thus enters the basin of a local minimum (or a very flat region) early, and the advantage of Newton-CG there comes from the path of its first steps rather than from its fast local convergence.

Finer grids. Refining the parameter grid to $m_p = 51$ or the forward grid to $m = 151$ does not change the picture (Table 6). Newton-CG remains about twice as fast (speedups from 1.3 to more than 2.5), and its advantage does not grow with the number of unknowns. An L-BFGS iteration costs one gradient for any m_p, while Newton-CG needs more Newton iterations at $m_p = 51$ (28 against 13 for tikhonov-grad with $\beta = 10^4$). At $m = 151$ the discretization error in the data is smaller, and the error in $\mathbf{a}$ drops from 0.09–0.12 to 0.08–0.10. At $m_p = 51$ it does not change, since it is set by the discretization error and not by the parameter grid.

In summary, with high-resolution data Newton-CG reaches the regularized minimizer about 1.5–4 times faster than L-BFGS at the same accuracy in $\mathbf{a}$, provided it does not stall at the initial guess, and starting it after about 20 L-BFGS iterations removes the stall. It does not give a better $\mathbf{a}$: both solvers find the same $\mathbf{a}$ on the resolved nodes, and its accuracy is limited by the discretization error in the data.

Case		error in **a**	iter.	$t_{10^{-4}}$ (h)	speedup
tikhonov-grad, $\beta = 10^3$, $m = 76$	L-BFGS	0.085	200	1.04	
	Newton-CG	0.088	21	0.63	1.6
tikhonov-grad, $\beta = 10^4$, $m = 76$	L-BFGS	0.073	200	>1.53	
	Newton-CG	0.086	13	0.35	>4.4
TV, $\beta = 100$, $\varepsilon = 10^{-3}$, $m = 76$	L-BFGS	0.104	200	1.63	
	Newton-CG	0.104	19	1.06	1.5
TV, $\beta = 1$, $\varepsilon = 10^{-4}$, $m = 251$	L-BFGS	0.094	200	10.39	
	Newton-CG	0.091	24	6.73	1.5

Table 4: High-resolution data, $m_p = 26$. Relative error in **a** *on nodes 6–25, iterations, wall time to $(J - J^*)/(J_0 - J^*) = 10^{-4}$, and speedup of Newton-CG over L-BFGS. A lower bound (>) means that L-BFGS did not reach the level.*

R, β	L-BFGS	Newton-CG	hybrid, $k = 20$	hybrid, $k = 40$
tikhonov-grad, $\beta = 10^4$	>1.53	**0.35**	0.57	0.92
TV, $\beta = 10$, $\varepsilon = 10^{-4}$	>1.52	1.89	**0.98**	1.75
TV, $\beta = 100$, $\varepsilon = 10^{-4}$	>1.46	6.47	**1.39**	1.39

Table 5: Hybrid L-BFGS/Newton-CG at $m = 76$: wall time (h) to $(J - J^*)/(J_0 - J^*) = 10^{-4}$, with J^* the smallest objective of the four runs. The hybrid starts Newton-CG from L-BFGS iteration k, and its time includes the L-BFGS part. A lower bound (>) means that the level was not reached within 200 iterations. Fastest in bold.

			error in **a**		$t_{10^{-4}}$ (h)		
m	m_p	R, β	L-BFGS	Newton-CG	L-BFGS	Newton-CG	speedup
76	26	tikhonov-grad, 10^3	**0.113**	0.115	1.04	0.63	1.6
151	26	tikhonov-grad, 10^3	**0.082**	0.177	2.50	>3.52	–
76	26	tikhonov-grad, 10^4	**0.092**	0.099	>1.53	0.35	>4.4
76	51	tikhonov-grad, 10^4	**0.096**	0.099	>1.66	0.88	>1.9
151	26	tikhonov-grad, 10^4	**0.087**	0.094	>4.47	1.82	>2.5
151	26	TV, 1	0.088	0.088	2.45	1.83	1.3
76	51	TV, 10	0.105	**0.104**	>1.38	0.86	>1.6
151	26	TV, 10	**0.084**	0.100	>4.05	1.71	>2.4

Table 6: Finer parameter grid ($m_p = 51$) and forward grid ($m = 151$), with $\varepsilon = 10^{-3}$ for TV. The error is the relative L^2 error of the linearly interpolated **a** *against the true step on the resolved region $-6 \le x \le 9.2$, which can be compared across m_p; it differs from the nodal error in Table 4. Wall time to $(J - J^*)/(J_0 - J^*) = 10^{-4}$ and speedup of Newton-CG. A lower bound (>) means that the level was not reached within the run.*

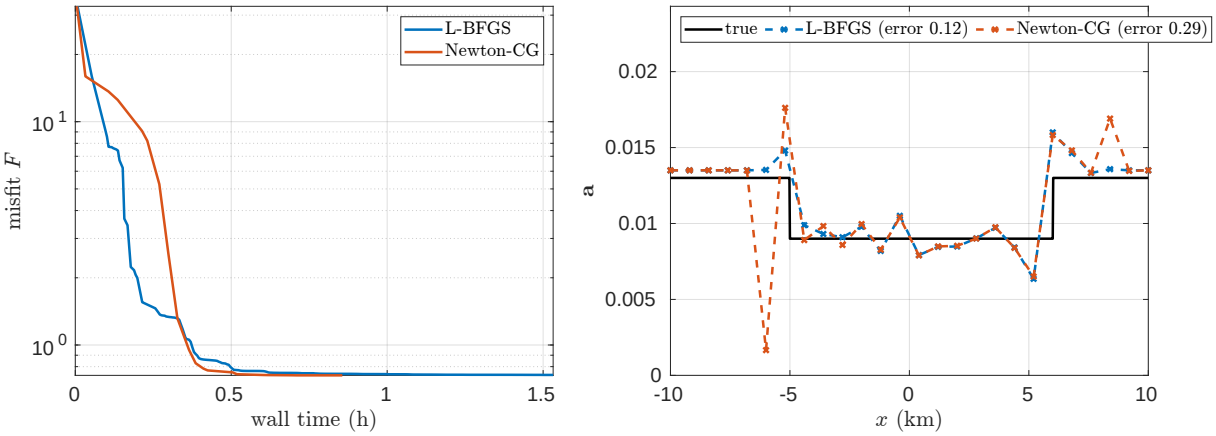


Figure 4: High-resolution data without regularization ($m = 76$). Left: misfit against wall time. Right: final **a** *with the relative error on nodes 6–25.*

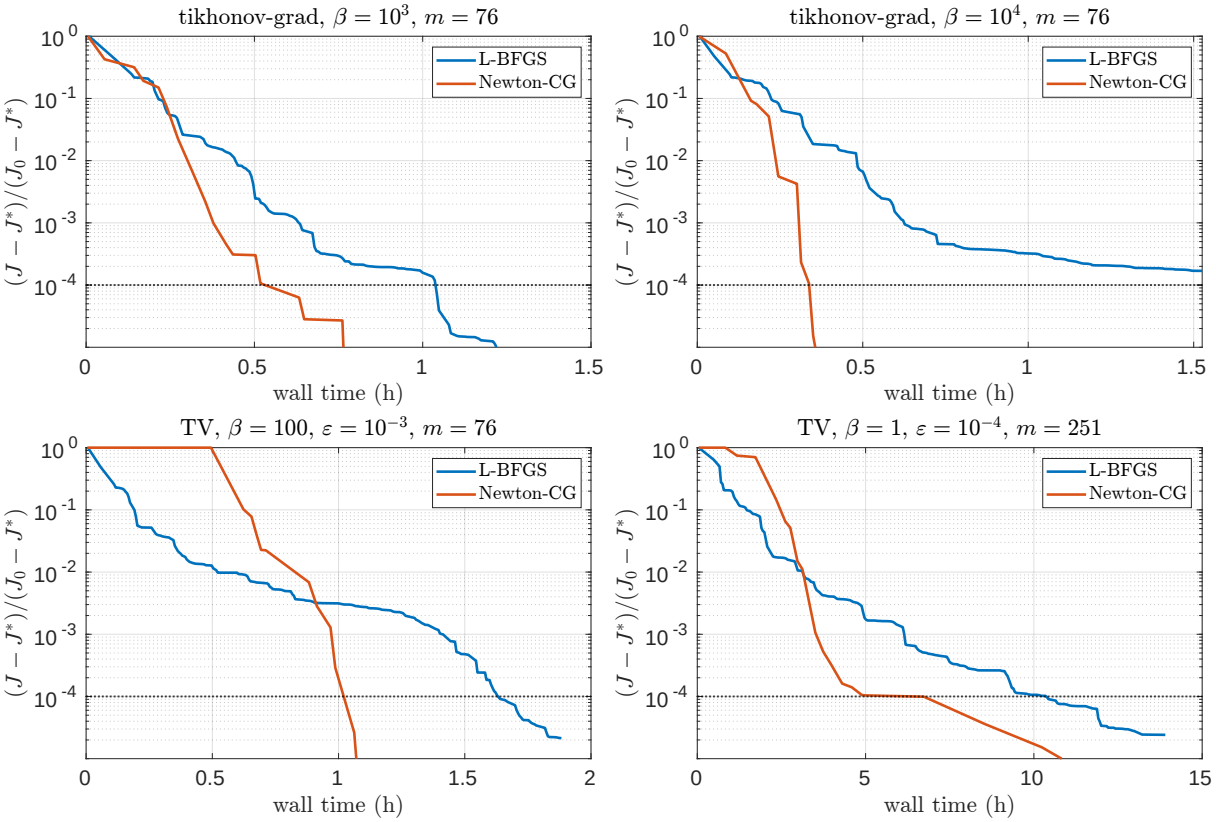


Figure 5: Relative objective against wall time for the cases of Table 4. The dotted line marks the level 10^{-4}.

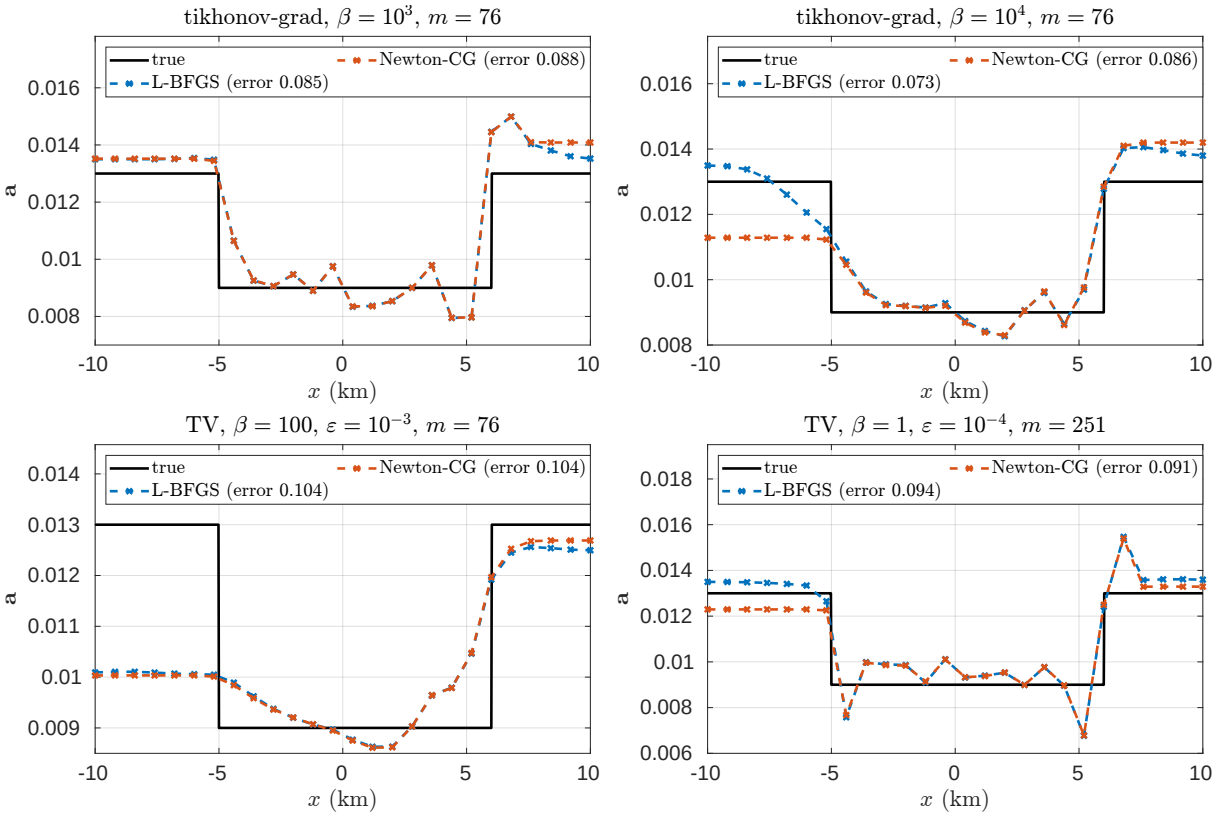


Figure 6: Final **a** *for the cases of Table 4, with the relative error on nodes 6–25. Nodes 1–5 and 26 are not resolved by the receivers.*

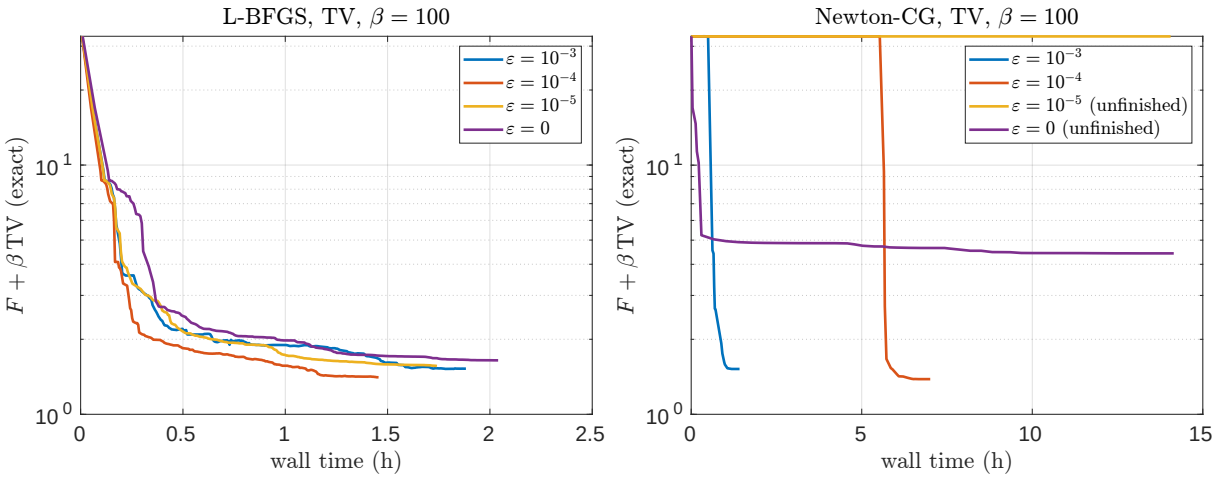


Figure 7: TV with $\beta = 100$ ($m = 76$) for different ε, evaluated on the exact objective $F + \beta\,\mathrm{TV}$. Runs marked unfinished were stopped before convergence.

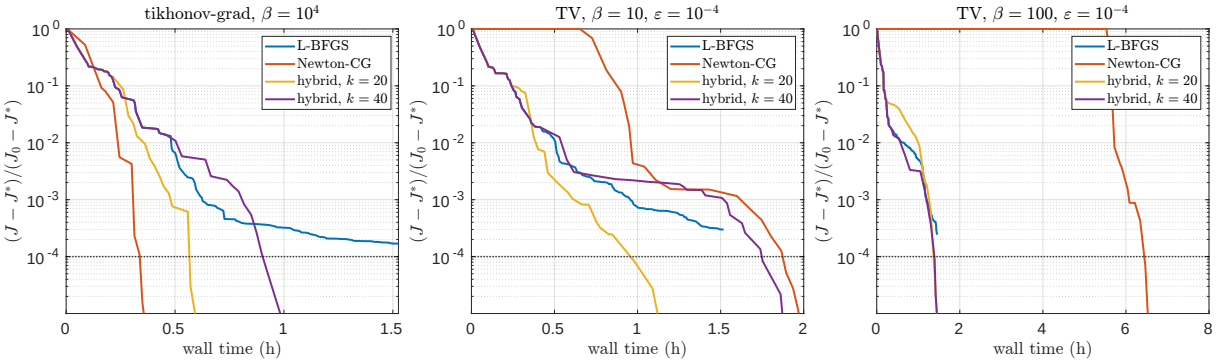


Figure 8: Relative objective against wall time for the hybrid L-BFGS/Newton-CG ($m = 76$). The hybrids follow the L-BFGS run until iteration k. The dotted line marks the level 10^{-4}.

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